Week 4 :
The Common Ground in the Global South

WI1103 Introduction of Sustainability Principles
(Week - 4)



4.1 : *Global North* and *Global South*

WI1103 Introduction of Sustainability Principles



“The Future We Want” (Rio+ 20s)
‘Realizing The Future We Want for All (2019)

- Response to 'Development' that increasingly 'damages the environment' and pays less attention to social and cultural dimensions.
- Raising the development approach in 'Global South' countries that is different from 'Global North'.



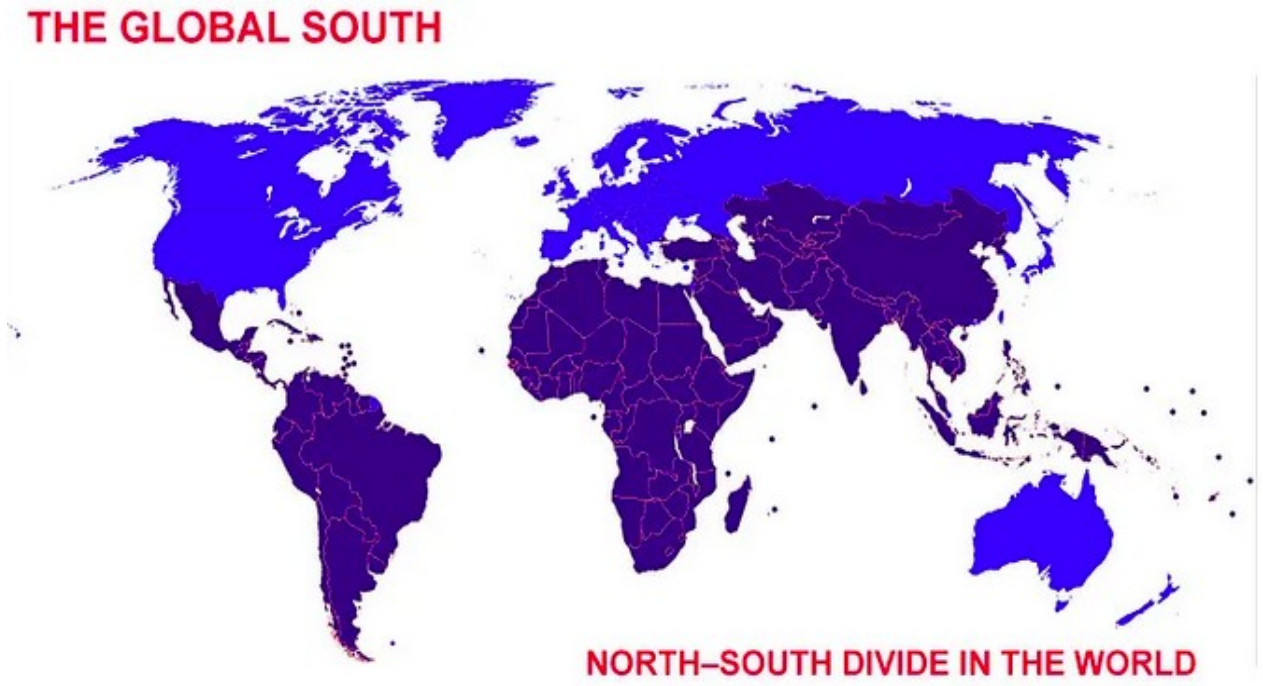
Based on week 3 lecture:

What increases environmental pressure?

- **Population growth** increases resource needs
- **Higher consumption** increases material and energy use
- **Technology** can reduce or actually increase environmental pressure
- Environmental pressure is not just about **how many resources are used**, but also how the **benefits and impacts are shared**
- The **distribution** of these benefits and impacts is **unequal across countries**. Therefore, we need to see why the **Global South** context becomes important in discussions on sustainability.

Why Different?

- Characteristic of Global South:
  - Economic development not as advanced as the Northern Hemisphere
  - 88% of the world's population
  - Larger land area
  - 80% middle class (2030)



Global North vs *Global South*

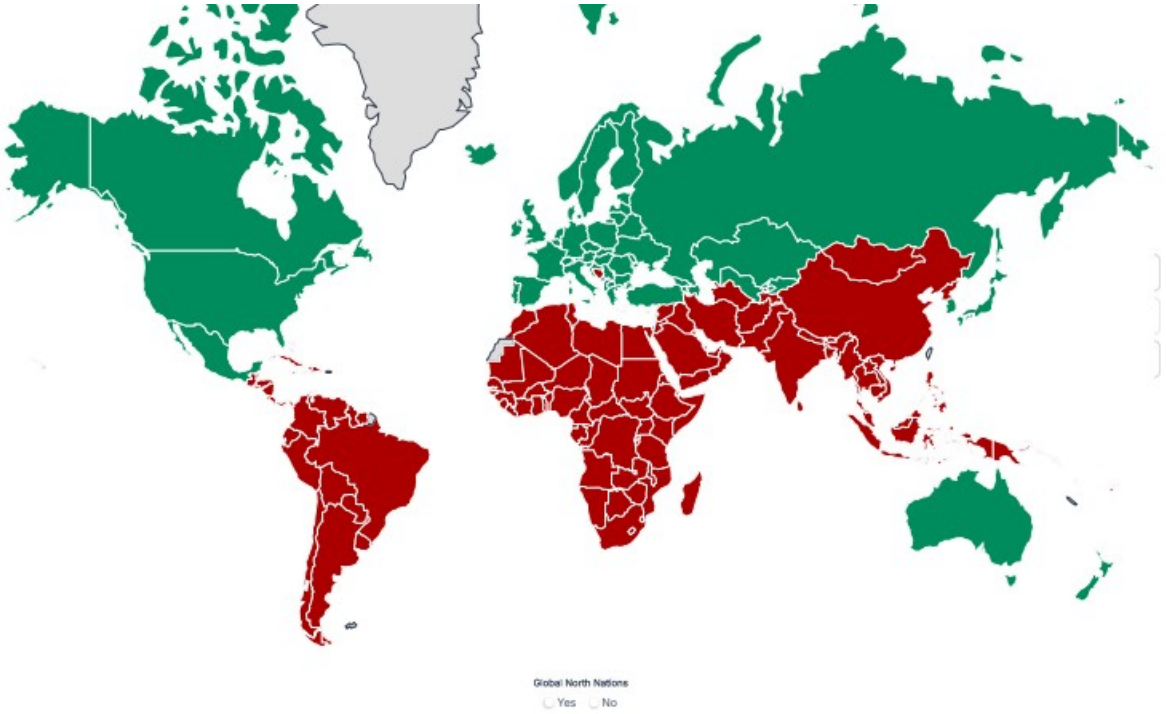
The category of countries between the Global South and North is not based on geographical location; rather, it is based on social and economic aspects.



List of Global North Country according to IMF

Country	Country	Country
Australia	Iceland	Portugal
Austria	Ireland	Puerto Rico
Belgium	Israel	San Marino
Canada	Italy	Singapore
Cypruss	Japan	Slovak Republic
Czech Republic	Korea	Slovenia
Denmark	Luxembourg	Spain
Estonia	Macao SAR	Sweden
Finland	Malta	Switzerland
France	Netherlands	Taiwan Province of China
Germany	New Zealand	United Kingdom
Greece	Norway	United States
Hong Kong SAR		

Countries of Global South



- India
- China
- Indonesia
- Pakistan
- Nigeria
- Brazil
- Bangladesh
- Ethiopia
- Kenya
- Myanmar
- Colombia
- Sudan
- Uganda
- Algeria
- Iraq
- Egypt
- Philippines
- DR Congo
- Vietnam
- Iran
- Tanzania
- Thailand
- South Africa
- Argentina
- Afghanistan
- Yemen
- Angola
- Morocco
- Mozambique
- Malaysia
- Ghana
- Saudi Arabia
- Peru
- Madagascar
- Ivory Coast
- Cameroon
- Nepal
- Niger
- Venezuela
- North Korea
- Syria
- Mali
- Burkina Faso
- Sri Lanka
- Malawi
- Zambia
- Chad

4.2 : *The commons* in the *Global South*

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Global South – The Commons

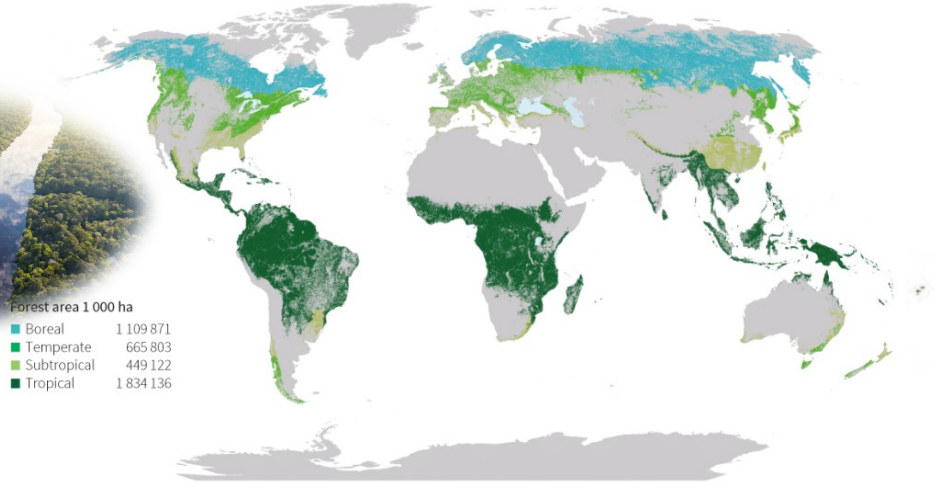
- The total number of countries categorized under the **Global South** is larger than the **Global North**.
- The **Global South** faces significant development needs alongside a high reliance on natural resources.
- Many resources in the **Global South** are common resources (**the commons**).

- ***Who uses them?***
- ***Who enjoys the benefits?***
- ***Who guards them to ensure they remain available?***





The global distribution of forests, by climatic domain



Source: Adapted from United Nations World map, 2020.

FAO. 2020. *Global Forest Resources Assessment 2020: Main report.* Rome.

The Commons Global South

- 'The commons' is a type of 'wealth' that is different (land, culture, values, activities, intangible heritage, etc.) that is 'owned and maintained' together.
- Often closely tied to cultural identity, spirituality, and the survival of indigenous and local communities.
- Example: Forests are not just timber; they are 'pharmacies', spiritual places, and guardians of the water cycle. Traditional knowledge is an intellectual 'commons'.
- Threats: Exploitation by corporations, government policies that do not involve local communities, climate change.







What are ‘The Commons’?

The Commons

- Water
- Air
- Land
- Forest
- Ocean
- Biodiversity
- Electromagnetic spectrum
- Outer space
- Etc.

Man Made The Commons

- Irrigation system
- Music
- Park
- Seeds
- Shared Knowledge
- Internet technology
- Bandwidth
- Library
- Etc.





4.3 : The importance of *The commons*

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The Importance of 'The Commons' - Ecosystem services

- Ecosystem services are everything provided by healthy and well-functioning ecosystems.
- According to the Millennium Ecosystem Assessment (2005), they consist of provisioning, regulating, supporting, and cultural services.
- Economic activities as a whole are intrinsically dependent on the sustainability of these ecosystem services.

The classification of ecosystem services according to the Millennium Ecosystem Assessment

PROVISIONING SERVICES	REGULATING SERVICES	CULTURAL SERVICES	SUPPORTING SERVICES
Products obtained from ecosystems	*Benefits obtained from regulation of ecosystem processes*	*Nonmaterial benefits obtained from ecosystems that enrich lives*	*Underpinning services that enable other services to function*
• Food • Fresh water • Fuel wood • Fibre • Biochemicals • Genetic resources	• Climate regulation • Disease regulation • Water regulation • Water purification • Pollination	• Spiritual & religious • Recreation & ecotourism • Aesthetic • Inspirational • Educational • Sense of place • Cultural heritage	• Soil formation • Nutrient cycling • Primary production
 (a)	 (b)	 (c)	 (d)

Source: modified from MEA (2005). Photos: (a), (c) and (d): © Mauro Masiero; (b) © Sepul Kanti Barua.

The Examples of Ecosystem Services

Natural Capital

Marine Ecosystems

Ecosystem Services	Economic Services
Oxygen supplied through photosynthesis	Food
Water purification	Energy from waves and tides
Climate moderation	Pharmaceuticals
CO_2 absorption	Harbors and transportation routes
Nutrient cycling	Recreation and tourism
Reduced storm damage (mangroves, barrier islands, coastal wetlands)	Employment
Biodiversity: species and habitats	Minerals





Top: Willyam Bradberry/Shutterstock.com. Bottom: James A. Harris/Shutterstock.com.

Natural Capital

Freshwater Systems

Ecosystem Services	Economic Services
Climate moderation	Food
Nutrient cycling	Drinking water
Waste treatment	Irrigation water
Flood control	Hydroelectricity
Groundwater recharge	Transportation corridors
Habitats for many species	Recreation
Genetic resources and biodiversity	Employment
Scientific information	





Top: Galyna Andrushko/Shutterstock.com. Bottom: Kletr/Shutterstock.com.

The Examples of Ecosystem Services

Natural Capital

Ecosystem Services of Rivers

- Deliver nutrients to sea to help sustain coastal fisheries
- Deposit silt that maintains deltas
- Purify water
- Renew and renourish wetlands
- Provide habitats for wildlife

Natural Capital

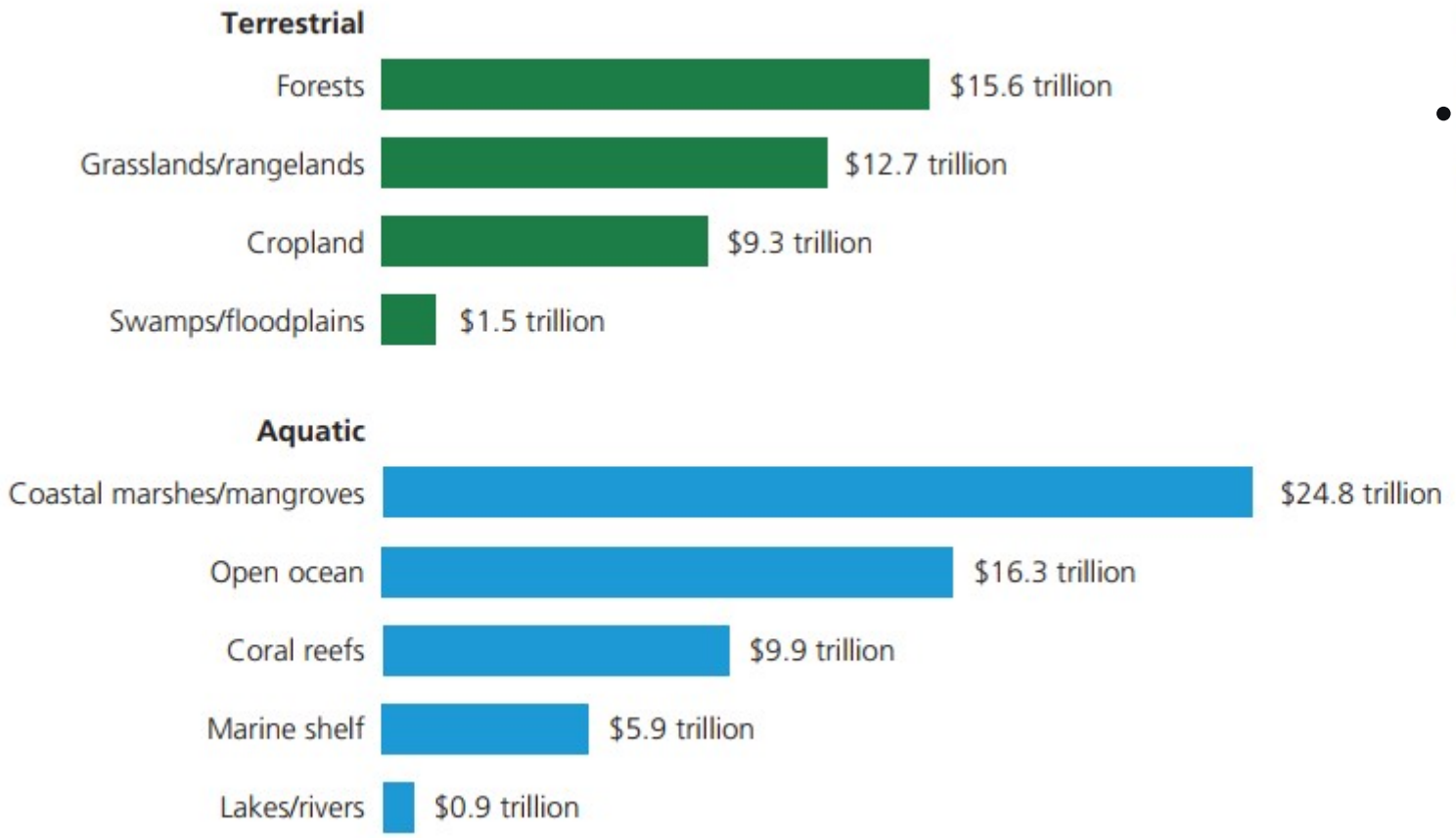
Forests

Ecosystem Services	Economic Services
Support energy flow and chemical cycling	Fuelwood
Reduce soil erosion	Lumber
Absorb and release water	Pulp to make paper
Purify water and air	Mining
Influence local and regional climate	Livestock grazing
Store atmospheric carbon	Recreation
Provide numerous wildlife habitats	Jobs



Photo: Val Thoermer/Shutterstock.com

Economic value estimation of ecosystem services



- The estimated monetary value of the ecosystem services provided each year by the world's major terrestrial and aquatic ecosystems (Spoolman and Miller, 2021).

How ecosystems services can be paid?

Economically: Payment for Ecosystem Services (PES)—often referred to as Environmental Service Payments (PJL):

a mechanism where parties benefiting from nature (e.g., clean water, healthier air, carbon absorption, scenic beauty) pay those who preserve/enhance those services (e.g., forest farmers, coastal communities), provided that the results are genuinely achieved.

Example: Cidanau, Banten

Socially: 'mutual Consent' or a joint agreement, as practiced by indigenous communities.

In terms of governance: government regulations.



4.4 : Example case of regulating 'The Commons'

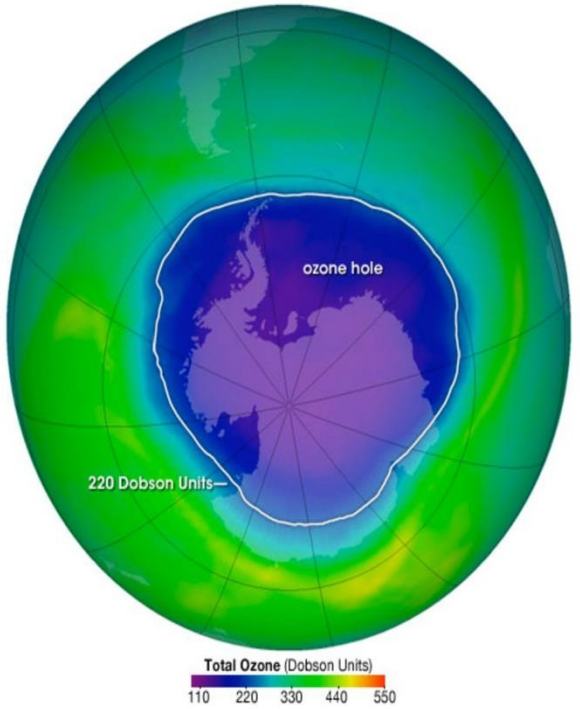
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Example: Ozone Hole in Antarctica

One of the Common Assets is the layer of air (ozone) that blocks harmful ultraviolet radiation from entering the Earth.

In 1985, the ozone layer thinned – 'hole' in the ozone over Antarctica.



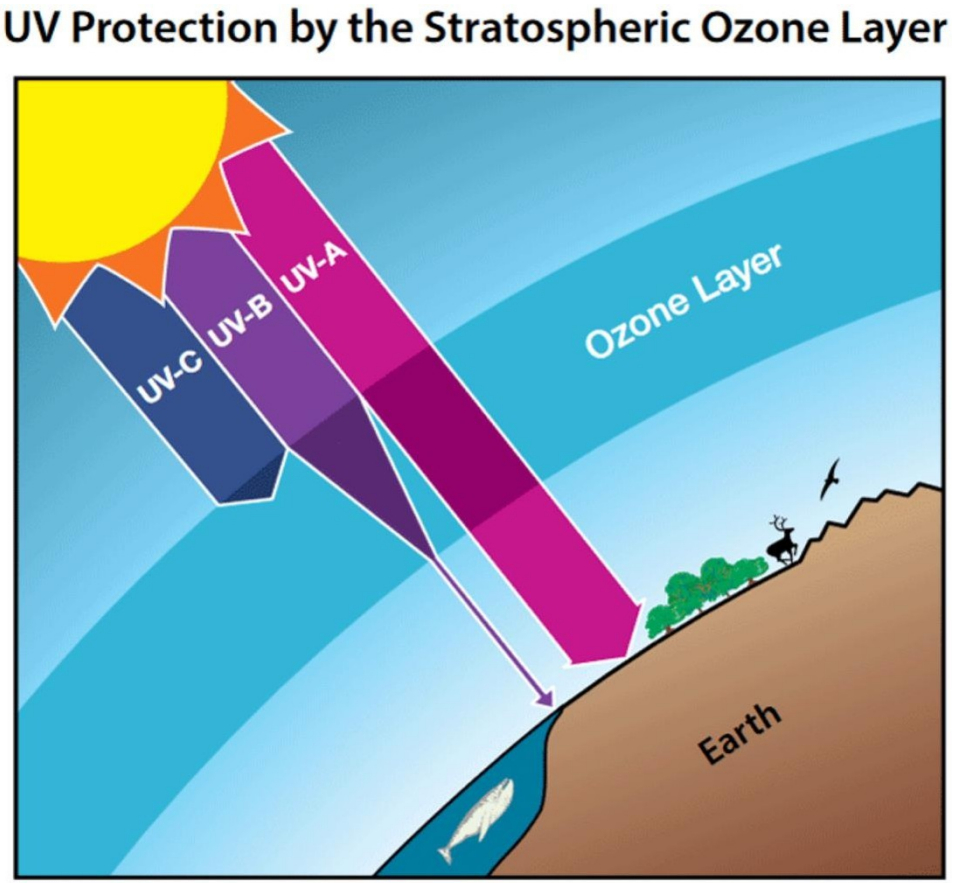
The ozone hole is the region over Antarctica with total ozone of 220 Dobson Units or lower. This map shows the ozone hole on October 4, 2004. The data were acquired by the Ozone Monitoring Instrument on NASA's Aura satellite.

The Danger of Ozone Layer Depletion & Regulation

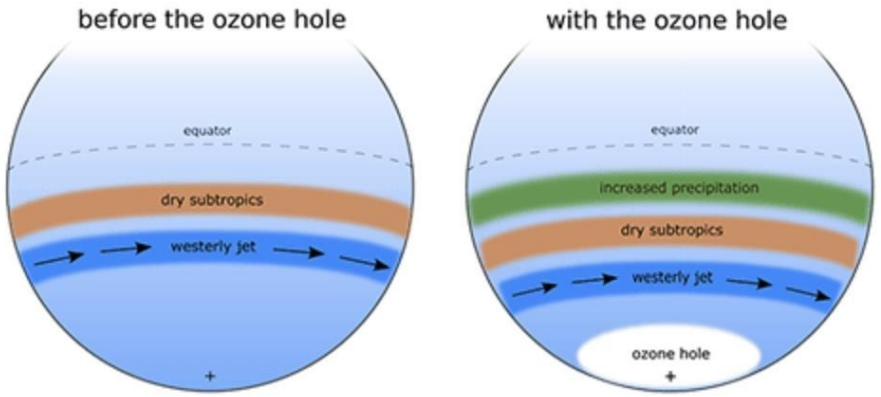
Endangering the survival of human habitats ~ initiating the existence of the Global Consensus

Montreal Protocol (1987): a global agreement prohibiting the production and use of hazardous substances (CFC and its derivatives) in the air ~

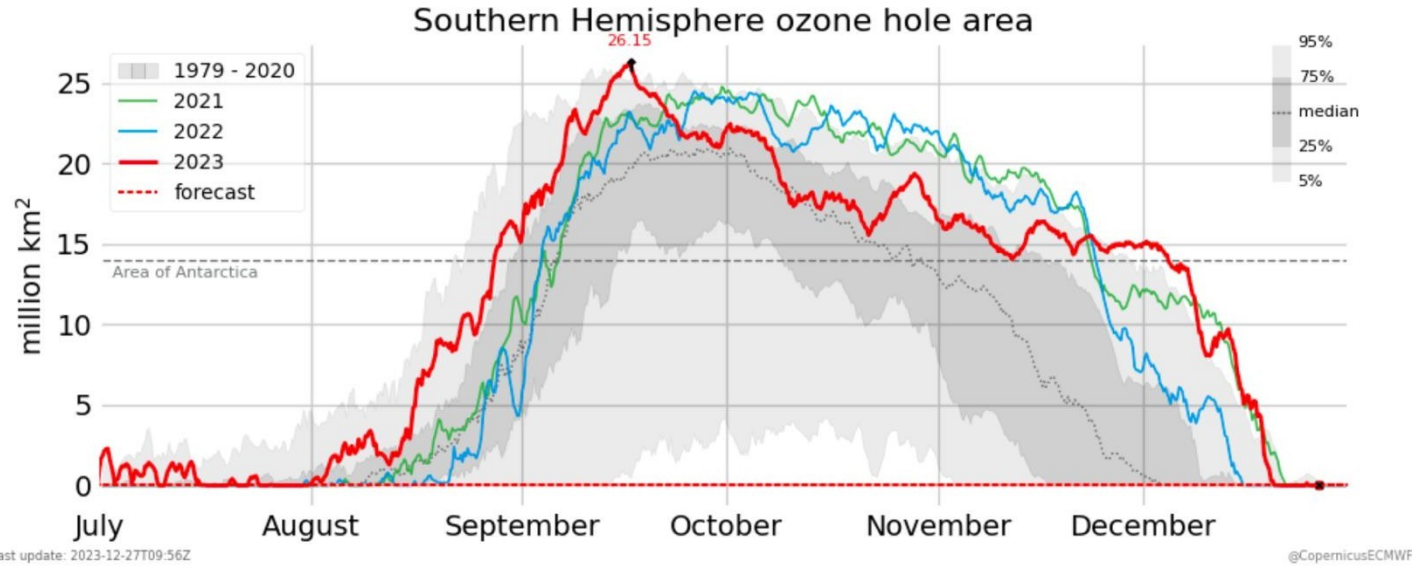
Kigali Amendment (2016) prohibits HFC, HCFC, and similar substances.



Monitoring Ozone Depletion: Southern Hemisphere



Depletion of ozone around the South Pole has altered global air-circulation patterns, bringing more rain to the subtropics. (Kang et al., 2011)



Southern Hemisphere 2023 ozone hole area compared with previous years and the 1979-2020 average. Credit: CAMS

"Common Property": Regulations Respected Together

The community gathers/works together to establish 'rules of the game' that can maintain the existence of 'common property'.

Example: Maintaining waterways.

Logic of Collective Action.



Example : Subak Irrigation System

Based on the **commons** concept, Subak fulfills the following key elements:

- **Shared resource:** water from the same source and stream is used by many farmers.
- **Multiple users:** a single water source can be managed jointly by farmer groups; UNESCO notes that around 50–400 farmers can manage the water supply from one source.
- **Collective rules:** the community collectively determines the allocation and timing of water use.
- **Collective management:** canals, dams, water temples, and distribution systems are maintained collectively.
- **Social-cultural dimension:** water management is not merely a matter of agricultural production, but is connected to rituals, local institutions, and *Tri Hita Karana*.



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Example: ‘Dieselgate’

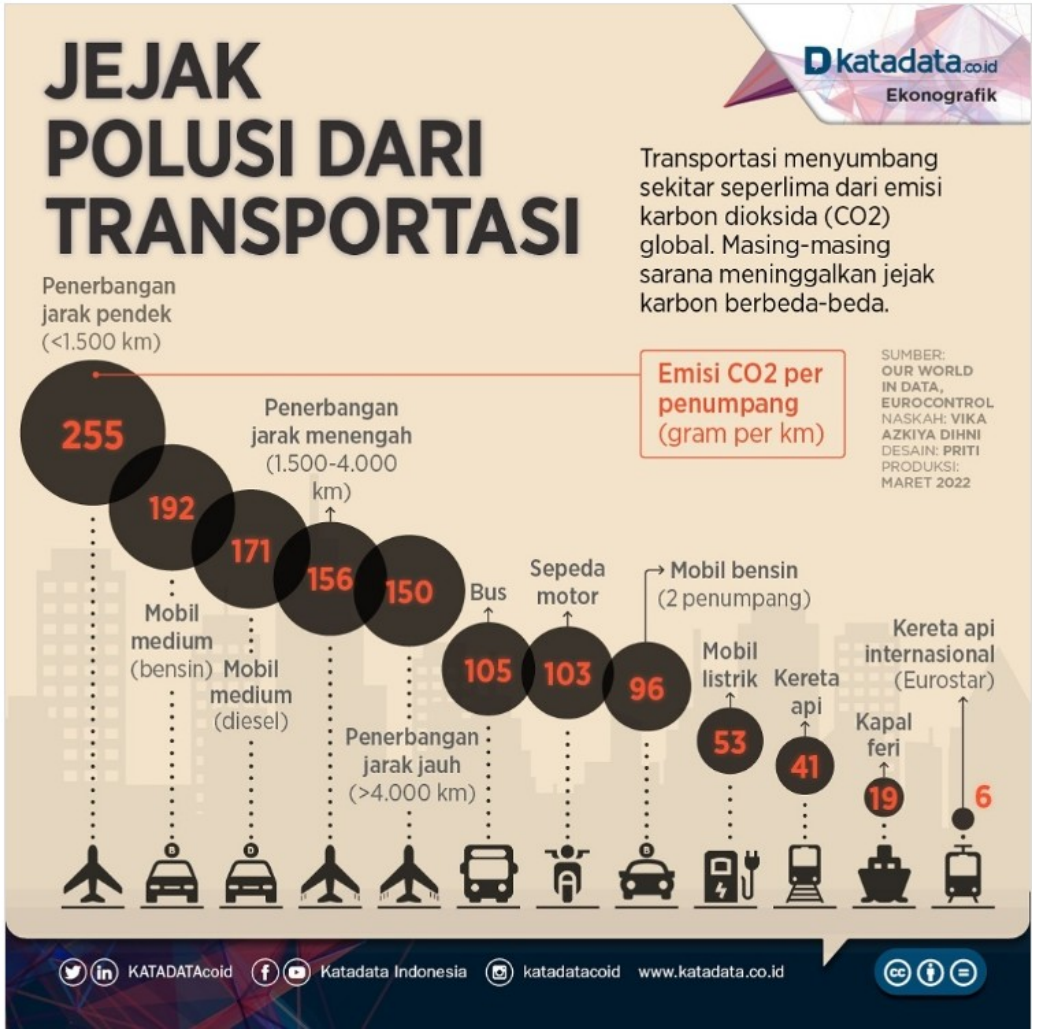
- Motor vehicle emissions in the US are regulated by the Clean Air Act.
- Strict regulations on NOx.
- VW campaigned ~ Diesel Engine 'clean technology’.
- Proven: The use of 'software' that could control emissions during testing & emissions increased 20x on the road.
- NOx emissions became 'smog' (health issue). Millions of vehicles were recalled from circulation.
- How is the implementation of vehicle emission reduction in Indonesia? (Global south)



Air pollution level regulation: Vehicle gas emission

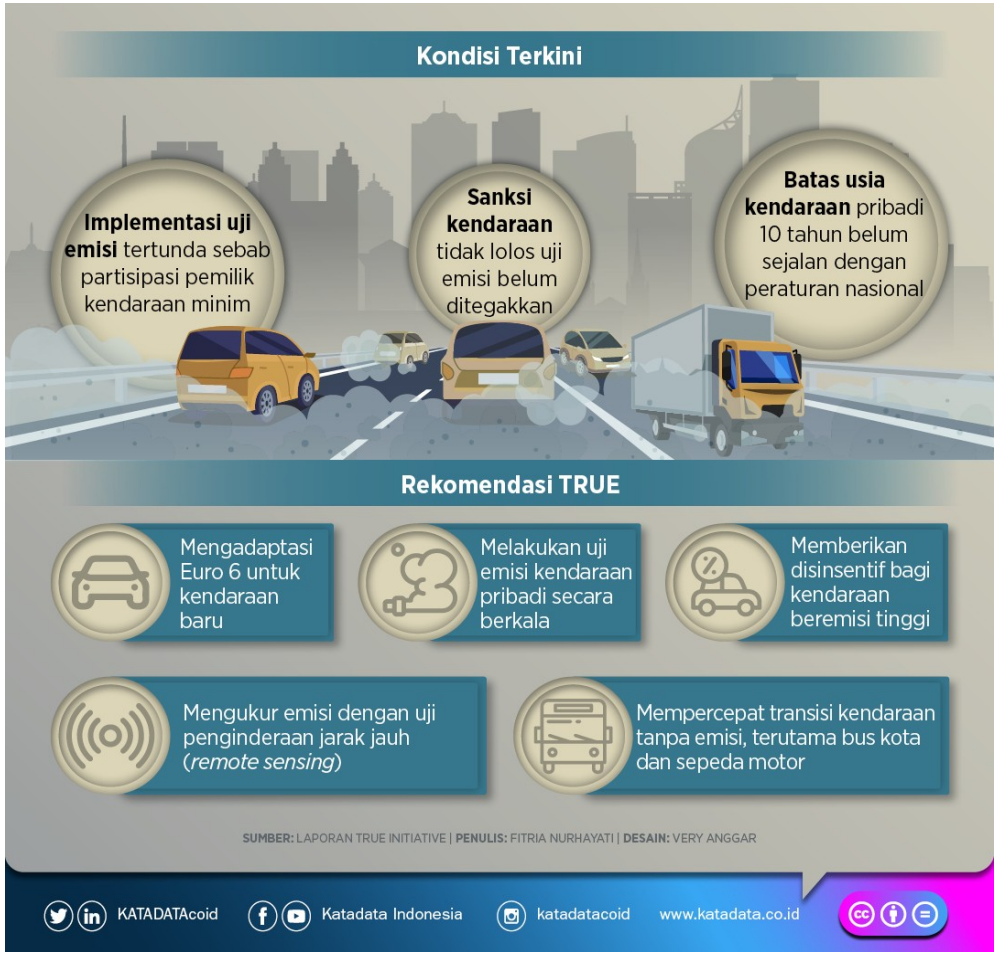
How is the regulation of vehicle exhaust emissions in Indonesia?

The regulation of vehicle gas emissions in Indonesia currently refers to the Minister of Environment and Forestry Regulation (Permen LHK) No. 8 of 2023 and Government Regulation (PP) No. 22 of 2021.



The regulation of vehicle gas emissions in Jakarta






1975
1990
2005
2011
2013
2015
2016
2019
2050
2060
US CAFÉ* implementation (FE concern)
FE: 7,6 km/L
EU start CO2 discussion
Stringent environment Policy (Euro 4)
US CAFE
FE: 12,8 km/L
FE: 16,5 km/L
CO2: 140 g/km
Global Trend
Indonesia Trend
GREEN MOBILITY
Current IDN
NET-ZERO
Narrowing GAP w/ Global Trend
(Market: 1.062 K)
(Market: 1.230 K)
Fuel subsidy taken out
(Market: 534 K)
(Market: 72 K)
LCEV
Trigger factor
• Energy security
• Environment
HYBRID
LCGC
Trigger factor
• Fuel consumption
• Low buying power
FE: 12,8 km/L
CO2: 180 g/km
EV
Initiated by 2-wheels vehicles & Transportation Public
CV based
Euro2 w/ CV based Platform

Policies & Challenges of Regulating Vehicle Gas Emissions in Indonesia

Main Policy:

- Euro 4 standards for new vehicles (mandatory compliance by manufacturers).
- Periodic emission testing (a requirement for vehicle tax renewal, regulated by Ministry of Environment and Forestry).
- Availability of clean fuel.
- Implementation

Challenges:

- Technical: Limited emission testing equipment, older vehicles struggle to meet standards.
- Economic: High repair costs, burden on public transport operators.
- Regulatory: Weak coordination among agencies, lax sanctions.
- Social: Low public awareness, enforcement outside Jakarta remains weak..

Air Quality in Bandung

IQAir

Data Kualitas Udara | Belajar | Berita | Dampak | D

Negara, kota, atau lokasi Anda...

Prakiraan per jam

Prakiraan indeks kualitas udara (AQI*) Bandung

Sekar...	13.00	14.00	15.00	16.00	17.00	18.00	19.00	20.00	21.00	22.00	23.00	Jum	00.00	01.00	02.00	03.00	04.00
173	170	168	167	167	162	156	148	138	126	112	113		116	125	121	117	114
31°	31°	30°	29°	28°	27°	25°	24°	23°	23°	22°	22°		22°	21°	21°	21°	21°
10 km/h	14 km/h	18 km/h	18 km/h	13 km/h	7 km/h	5 km/h	4 km/h	3 km/h	3 km/h	2 km/h	2 km/h		1 km/h	1 km/h	3 km/h	3 km/h	4 km/h
37%	35%	35%	41%	48%	53%	58%	62%	67%	70%	73%	76%		79%	80%	81%	82%	83%

Dunia / Indonesia / West Java / **Bandung**

Kualitas udara di Bandung

Indeks kualitas udara (AQI*) dan polusi udara PM2.5 di Bandung • 469.2K Pengikut • 12.00, Sep 10

Atribusi data

5 stasiun dari **3 contributors**

Prakiraan setiap hari

Prakiraan indeks kualitas udara (AQI*) Bandung

Hari ini	136	31°	21°	18 km/h	56%
Jum	101	34°	20°	23 km/h	52%
Sab	119	33°	21°	21 km/h	57%
Min	91	31°	20°	17 km/h	56%
Sen	72	34°	19°	20 km/h	41%
Sel	75	33°	22°	13 km/h	50%
Rab	78	34°	22°	16 km/h	52%

Polutan udara

Berapa kualitas udara saat ini di dekat Bandung?

PM2.5

Fine particles (≤ 2.5 µm)

87 µg/m³

Konsentrasi PM2.5 saat ini **17.4** kali nilai panduan PM2.5 tahunan WHO.

173 AQI US

Tidak sehat

Polutan utama: PM2.5

87 µg/m³

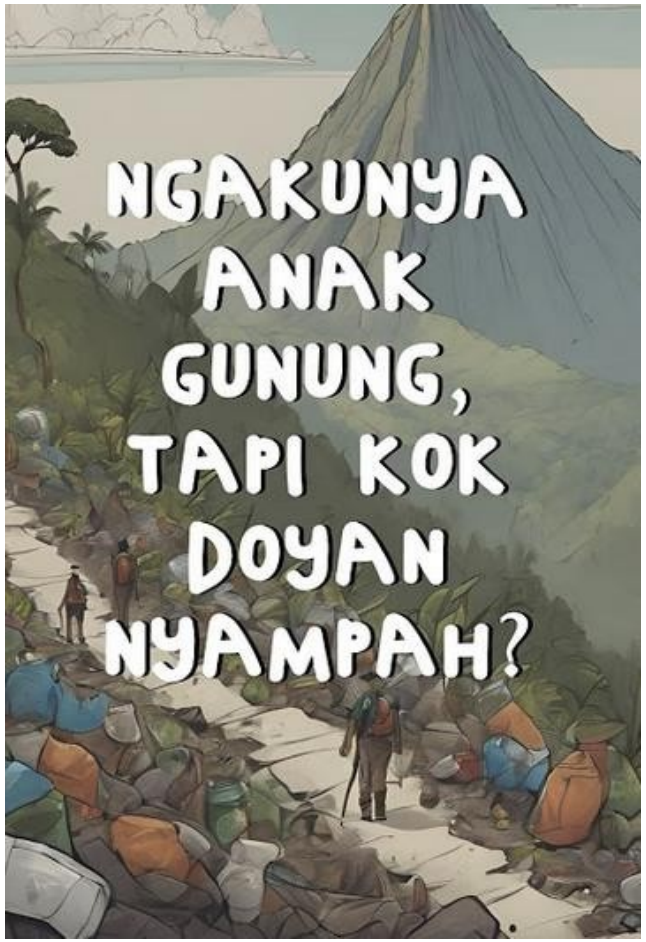
31°

10 km/h

37 %

How is the arrangement made?

- Boundaries
- Costs
- Opportunities for Use
- Supervision
- Sanctions
- Conflict Resolution



Technological Progress and ‘The Commons'

- Technological advancement can drive changes in the management of 'common property' (the commons), which in some cases leads to exclusion or reduced joint access, although in other cases it actually strengthens collective capacity to manage it.





- Fence / Stake
- Information and data
- Energy
- Time-saving technology (automation, smart gadgets)
- (Intellectual) Property Rights

“The commons” with non-physical boundaries

- Many boundaries are non-physical, making them invisible or intangible.
  - *Example:* Air (cannot be bounded); Waters (boundaries are not clearly visible)
  - Air, oceans, groundwater, fish stocks, and knowledge can move or be used across territorial boundaries.
- The primary function of a *boundary* is not just to erect physical barriers, but to define rights and limits on the usage of common resources.
- Technology helps make invisible boundaries measurable and monitorable, for example through: satellites, GPS, sensors, environmental quality data, and digital permits.

Example: Maritime Boundaries

- No physical fences mark boundaries in the ocean.
- UNCLOS establishes jurisdictional boundaries legally.
- Boundaries are defined as geographic coordinates.

Resource use rights differ according to zones:

- Territorial sea: $\leq$ 12nautical miles
- EEZ (Exclusive Economic Zone): $\leq$ 200 nautical miles
- **High seas:** beyond national jurisdiction



Discussion

1. Find and analyze one example of successful commons resource management. What factors influence the success of that management?
2. Identify a commons resource management issue in your surroundings. Based on what you learned from the chosen case, what can be implemented for commons resource management in your area.
3. Evaluate the potential challenges of that management (you can look at factors such as community characteristics, resources, and external support that can influence the success of the management).

Thank you